Dynamic Stability and Levitation Control in Maglev Systems

We present an experimental, project-based approach to teaching classical electrodynamics through the construction and analysis of a scaled-down magnetic levitation train. Using flexible magnetic tape as a guideway, we demonstrate stable levitation and perform precise measurements of the vertical gap as a function of load. These experimental data are validated using an analytical force model derived from a dipole-layer approximation, which confirms the *self-restoring* vertical stability of the repulsive configuration. By bridging laboratory measurements with theoretical energy methods, students can extrapolate their scaled results to industrial parameters, successfully reproducing the passenger capacity of a full-scale commercial maglev vehicle. This project offers a tangible platform for applying core concepts such as magnetic field geometry, vector potentials, and Earnshaw's theorem to a high-interest, real-world engineering challenge.

I. INTRODUCTION

Imagine a toy train floating effortlessly above its track, defying gravity and gliding smoothly with minimal friction. This captivating phenomenon is the essence of magnetic Levitation (Maglev) technology, which has the potential to revolutionize transportation by enabling high-speed, low-friction travel[1]. Maglev trains operate by using magnetic forces to lift and propel vehicles along a guideway, eliminating the need for traditional wheels and rails. This not only reduces mechanical wear but also allows for higher speeds and smoother rides.

A major challenge in Maglev design is achieving stable levitation while maintaining safety and efficiency under varying load conditions. To successfully meet these operational requirements, two distinct Maglev system approaches were conceived: Electromagnetic Suspension (EMS), which relies on attractive magnetic forces for levitation, and Electrodynamic Suspension (EDS), which uses repulsive forces generated by currents induced in conductive guideways[2,3]. EMS systems require active control to counteract inherent instabilities dictated by Earnshaw's Theorem[4], but can operate at any speed, whereas EDS systems are passively stable but require minimum speed for levitation.

In this work, we explore the physics of a simplified magnetic levitation system using repulsive magnet–magnet interactions. While commercial EMS trains rely on attractive electromagnetic forces with active feedback control, our experimental system uses repulsive permanent magnets arranged in a monopolar configuration. This allows us to isolate and analyze the static magnetic force–gap relation without the additional complexity of active stabilization. A key aim is to demonstrate how fundamental concepts from classical electrodynamics, typically covered in an undergraduate curriculum[5,6], govern the dynamics and performance of Maglev systems. For this analysis, we focus primarily on the inherent magnetic dynamics, treating active stabilization (required in practice and implemented via feedback control in commercial applications) as beyond the scope of this work[1]. We derive an analytical relationship between the levitation gap and the mass added to the train using a magnetic dipole approximation and then corroborate the theoretical results with measurements from a scaled-down experimental train. Application of the model to a full-scale train is finally discussed. By comparing theory and experiment, we show how even a reduced-scale system can capture the essential physics of full-scale Maglev trains, making this approach a valuable tool for teaching core electromagnetic principles in a tangible, real-world context. Understanding this load-gap relationship is essential for optimizing passenger capacity in commercial Maglev systems.

II. EXPERIMENTAL SETUP

II.1. The Scaled-Down Maglev Train Experiment

A scaled-down Maglev train model was constructed to investigate how the levitation height of the train changes with added mass. The train's base was a wooden block with dimensions of 12 cm (length), 5 cm (width), and 2 cm (height), fitted with Neodymium magnets on its underside. These magnets interacted with magnetic tape laid on the track to achieve levitation. The total mass of the train, including the magnets and the wooden block, was $m_0 = 62 \pm 1$ g. The uncertainty in mass measurements is attributed to the precision of the weighing scale used.

The track consisted of a 50 cm strip of magnetic tape attached to a cardboard base, with a width of 8 cm. To maintain lateral stability, 90-degree angle pieces were affixed to the edges of the track, preventing the train from sliding during experimentation. A key component of the setup was the use of monopolar magnetic tape. Based on the principle employed in the bonded Nd–Fe–B magnet fabrication demonstrated by Nlebedim et al.[7], where strong external magnetic fields are applied to induce internal domain alignment, we repolarized commercially available bipolar magnetic tape. This was achieved by rubbing it with neodymium magnets. This method effectively realigns the tape's magnetic domains to form a monopolar strip as required for stable levitation.

The experimental procedure involved gluing two strips of repolarized magnetic tape onto the underside of the train to form the levitating mechanism. Similarly, two long strips of magnetic tape were affixed to the track to act as rails, with the same monopolar alignment. This

configuration ensured stable levitation for the experiment.

The magnetic tape used in this experiment is a permanently magnetized material and does not generally require re-magnetization between trials. However, repeated mechanical contact, abrasion, or exposure to strong opposing fields can locally reduce or alter the magnetization. For reproducibility, the rail polarity and qualitative field strength should be verified prior to each experimental run (e.g., using a compass or a Hall sensor) to ensure that the magnetization pattern remains uniform along the track.

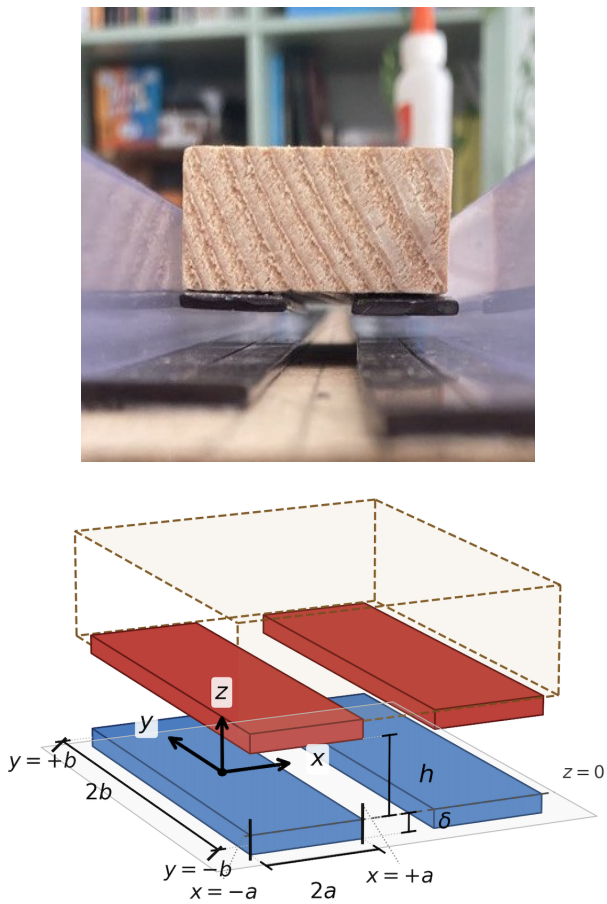

Figure 1. Top: Scaled-down experimental maglev train showing wooden base with neodymium magnets and cardboard cup for additional weights. Bottom: Schematic diagram of experimental geometry showing rail dimensions ($2a \times 2b \times \delta$), track width, and levitation height measurement setup. $2a = 1.25$ cm, $2b = 12.5$ cm, $\delta = 0.8$ mm.

II.2. Measurement of Levitation Height

To study the effect of mass on levitation height, a lightweight cardboard cup was placed on top of the train to hold additional weights. Rice was used as an adjustable weight source, allowing precise increments of mass to be added. The mass of the system was measured using a weighing scale with a precision of 1 g, leading to a measurement uncertainty of ± 1 g. The levitation gap (h) was measured using a ruler with a resolution of 0.5 mm, so the gap measurement uncertainty is ± 0.5 mm.

The experimental data for levitation gap (h) and mass (m) are summarized in Table I (Appendix A). These measurements, shown and commented on below in Figure 4, formed the basis for further analysis.

III. MODELING THE LIFT

III.1. Force Balance

Maglev systems rely on electromagnetic forces to counteract gravitational forces and achieve suspension. The net force acting on the levitating body is defined as:

$$F_{\text{net}} = 2F_{\text{mag}}(h) - (m + m_0)g, \tag{1}$$

where $F_{\text{mag}}(h)$ represents the upward repulsive force generated by a single pair of magnets separated by a gap h. The downward gravitational force is determined by the total mass, where m_0 is the mass of the empty system, m is the added load, and g is the acceleration due to gravity.

To ensure equilibrium at a specific levitation height, the system must satisfy:

$$2F_{\text{mag}}(h) = (m + m_0)g. \tag{2}$$

The relationship between levitation height (h) and added mass (m) was derived by modeling the forces acting on the train. To simplify calculations, we assumed the lower and upper magnets were perfectly aligned, thereby neglecting the cross-coupling (diagonal) magnetic forces between opposing sides. The geometry used in this derivation is shown in Fig. 1.

III.2. Model for the vertical magnetic field

The magnetic force exerted by the guideway on the train in the vertical direction (z) was calculated by considering the magnetic dipole interaction. The magnets are modeled as a distribution of differential dipole moments, $d\boldsymbol{\mu}^{(l)} = +M dx dy dz\, \hat{\mathbf{z}}$, where M represents the uniform vertical magnetization of the lower magnet (denoted by the superscript (l)).

The magnetic vector potential $\mathbf{A}$ at a point $\mathbf{r} = (x, y, z)$ due to an infinitesimal dipole moment located at $\mathbf{r}' = (x', y', z')$ is given by:

$$d\mathbf{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \frac{d\boldsymbol{\mu}^{(l)} \times (\mathbf{r} - \mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|^3} \tag{3}$$

where μ_0 is the permeability of free space ($\mu_0 = 4\pi \times 10^{-7}\,\text{N/A}^2$).

To find the total vector potential generated by the lower rail of width $2a$ and length $2b$, we expressed $d\mathbf{A}(\mathbf{r})$ in Cartesian coordinates and performed the integration over the entire volume of the bottom rail. Evaluating the integrals as indicated in appendix B, we find

$$\mathbf{A}(\mathbf{r}) = B_0 \sum_{i,j,k=1}^{2} s_{ijk} \left[\mathcal{L}(X_i, Y_j, Z_k)\hat{\mathbf{x}} - \mathcal{L}(Y_j, X_i, Z_k)\hat{\mathbf{y}}\right] \tag{4}$$

where $B_0 = B_r/(4\pi)$ and $B_r = \mu_0 M$ is the remanence of a magnet with uniform magnetization M. The geometric terms are defined by the corner indices $s_{ijk} = (-1)^{i+j+k}$, with coordinates $X_i = x + (-1)^i a$, $Y_j = y + (-1)^j b$ and $Z_k = z + (k-1)\delta$ for $i, j \in \{1, 2\}$ and $k \in \{1, 2\}$. The function $\mathcal{L}(u, v, w)$ is defined as:

$$\mathcal{L}(u, v, w) = -v \arctan\left(\frac{wu}{vR(u, v, w)}\right) + u \operatorname{asinh}\left(\frac{w}{(u^2+v^2)^{1/2}}\right) + w \operatorname{asinh}\left(\frac{u}{(v^2+w^2)^{1/2}}\right),$$

with $R(u, v, w) = (u^2 + v^2 + w^2)^{1/2}$.

III.3. Magnetic field distribution

The magnetic field $\mathbf{B}$ is derived from the vector potential $\mathbf{A}$ using the relation: $\mathbf{B} = \nabla \times \mathbf{A}$.

A direct calculation from Eq. (4), detailed in Appendix B, yields the axial component of the magnetic field:

$$B_z = B_0 \sum_{i=1}^{2}\sum_{j=1}^{2}\sum_{k=1}^{2} s_{ijk} \arctan\left(\frac{Z_k R_{ijk}}{X_i Y_j}\right). \tag{5}$$

where $R_{ijk} \equiv R(X_i, Y_j, Z_k)$. This result is in agreement with the model of Akoun and Yonnet[8]. Furthermore, it takes the same form as the field of a cuboid solenoid derived by Hampton et al.[9], who obtained this expression by modeling the magnet via equivalent Amperian surface currents $\mathbf{K} = \mathbf{M} \times \hat{\mathbf{n}}$, where $\hat{\mathbf{n}}$ denotes the unit outward normal to the lateral surfaces of the magnet.

Figure 2 illustrates the spatial variation of the vertical magnetic field component, B_z, showing normalized transverse profiles across the rail width. As anticipated, the field intensity decays as the air gap height (h) increases. Furthermore, the sharp lateral drop in field strength observed for $|x| > a$ validates the assumption that interactions between diagonally positioned magnets can be neglected.

The analytical expression (5) for the vertical magnetic field can be interpreted geometrically using the concept of solid angle. Let $\Omega(x, y, z)$ denote the solid angle subtended by a rectangular surface of dimensions $2a \times 2b$ at an observation point (x, y, z), defined as:

$$\Omega(x, y, z) = \sum_{i=1}^{2}\sum_{j=1}^{2} (-1)^{i+j} \arctan\left(\frac{X_i Y_j}{z R_{ij}}\right), \tag{6}$$

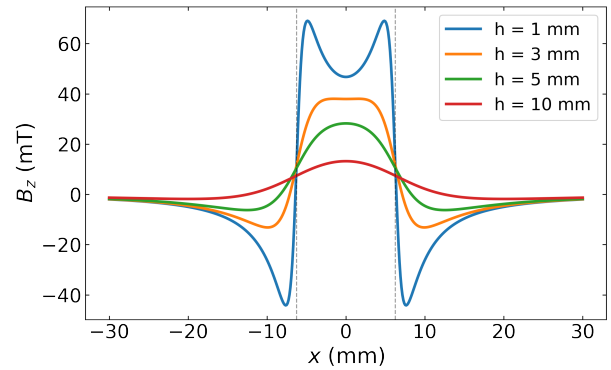

Figure 2. Magnetic field strength profiles $B_z(x, y = 0, z = h)$ along the transverse direction (x) of a rail. Here $B_r = 1.2$ T.

where $R_{ij} \equiv R(X_i, Y_j, z)$. By substituting this definition into the expression for B_z, the total field simplifies to the weighted difference between the solid angles subtended by the top and bottom faces of the magnet:

$$B_z(x, y, z) = \frac{B_r}{4\pi}\left[\Omega(x, y, z) - \Omega(x, y, z + \delta)\right]. \tag{7}$$

Physically, this representation confirms that the vertical field is determined by the net geometric flux originating from the equivalent magnetic pole sheets. As the gap z approaches zero, the solid angle of the top surface approaches 2π for points directly above the magnet, leading to the characteristic plateau in the field profile.

III.4. Analysis of the On-Axis Magnetic Field Limit

On the z-axis ($x = y = 0$), the symmetry of the cuboid magnet provides an intuitive interpretation of the field behavior through the lens of the solid angle. In this configuration, the observation point is centered directly above the magnetic pole, and the field is given by:

$$B_z(0, 0, z) = \frac{B_r}{4\pi}\left[\Omega_{\text{top}}(z) - \Omega_{\text{bottom}}(z + \delta)\right]. \tag{8}$$

In the near-field limit as the observation point approaches the center of the top surface ($z \to 0^+$), the top face effectively appears as an infinite plane. Geometrically, a point on the surface of an infinite sheet subtends a solid angle of 2π steradians (half of the total sphere). Since the rail magnet has a finite thickness δ, the contribution from the bottom face remains smaller than 2π, and for a sufficiently thick magnet ($\delta \gg z$), the surface field approaches:

$$\lim_{z \to 0^+} B_z(0, 0, z) \approx \frac{B_r}{4\pi}(2\pi) = \frac{B_r}{2}. \tag{9}$$

This confirms the classic physical result that the magnetic flux density at the surface of a large permanent magnet is exactly half of its remanence B_r.

Conversely, in the far-field limit ($z \gg a, b$), the rectangular faces appear as point sources. In this regime, the solid angle Ω scales with the inverse square of the distance (A/z^2), and the difference between the two terms $\Omega(z) - \Omega(z+\delta)$ scales as $1/z^3$. This transition demonstrates how the solid angle formulation elegantly captures the shift from a surface-pole behavior in the near-field to a characteristic dipole decay in the far-field.

III.5. Magnetic field lines

To further characterize the interaction between the two magnetized rails, we computed the magnetic field distribution in the vertical plane passing through the center of the system. Figure 3 shows the resulting field lines for a representative gap.

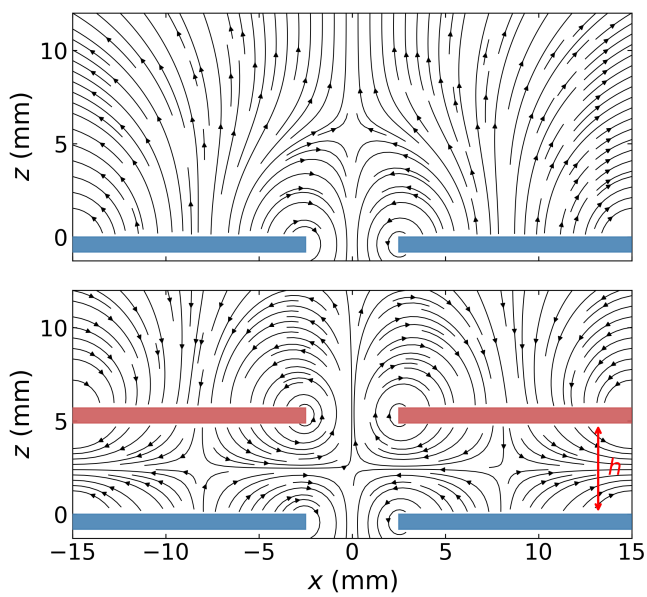

Figure 3. Magnetic field lines in the vertical plane ($y = 0$) for the two-rail experimental configuration. Top: lower track rails only (blue), showing the field produced by the guideway. Bottom: complete system with the upper train rails (red, magnetized downward) at the equilibrium gap $h \approx 4.9\,\mathrm{mm}$.

The field structure reveals strong confinement in the interaction region and rapid spatial decay, which directly explains the sensitivity of the lifting force to the gap h. This behavior is consistent with the shape of the potential energy well observed in Figure 6, and highlights the localized nature of the magnetic interaction responsible for levitation. The strong spatial gradient of the magnetic field implies a rapidly varying force with respect to the gap, which governs the vertical stiffness of the system and its stability properties.

III.6. Force on the upper rail

The upper rail, placed at a height z, consists of a volume of infinitesimal dipole moments aligned downward:

$$d\boldsymbol{\mu}^{(u)} = -M\,dx\,dy\,dz\hat{\mathbf{z}}.$$

The force on each dipole in the presence of a magnetic field is given by:

$$d\mathbf{F} = \nabla(d\boldsymbol{\mu}^{(u)} \cdot \mathbf{B}).$$

The force component along the z-axis is then expressed as

$$dF_z = -M\partial_z B_z(x,y,z)\,dx\,dy\,dz.$$

To find the total force due to a pair of rails, we integrate over the volume of the upper rail, as detailed in appendix B. The magnetic force $F_{\mathrm{mag}}(h)$ is expressed for a gap h as

$$F_{\mathrm{mag}}(h) = P_{\mathrm{mag}}\,\Delta_\delta^{(2)}\mathcal{S}(h). \tag{10}$$

where $P_{\mathrm{mag}} = \mu_0 M^2/\pi$ denotes a scaled magnetic pressure and the operator

$$\Delta_\delta^{(2)}\mathcal{S}(h) = \mathcal{S}(h+2\delta) - 2\mathcal{S}(h+\delta) + \mathcal{S}(h) \tag{11}$$

emphasizes that the total force results from the superposition of four distinct surface-to-surface interactions. Physically, it reflects the cancellation and reinforcement of fields between the four equivalent magnetic pole sheets of the rail and the levitating magnet. In the limit of a very small thickness ($\delta \to 0$), this finite difference converges toward the second derivative $\delta^2\mathcal{S}''(h)$, linking the discrete summation to the curvature of the geometric surface function $\mathcal{S}$, derived in detail in Appendix B:

$$\begin{aligned}\mathcal{S}(z) = 4ab\left[\arctan\left(\frac{4ab}{zR_{ab}}\right) - \frac{\pi}{2}\right] - z(R_{ab} - R_a - R_b + z)\\ - 2z\left[a\ln\left(\frac{(R_{ab}+2a)z}{(R_a+2a)R_b}\right) + b\ln\left(\frac{(R_{ab}+2b)z}{(R_b+2b)R_a}\right)\right],\end{aligned} \tag{12}$$

where $R_{ab} = \sqrt{(2a)^2 + (2b)^2 + z^2}$ and $R_{a,b}$ denote the distances evaluated at the respective edges $R(2a, 0, z)$ and $R(0, 2b, z)$.

In the limit $h \to \infty$, an asymptotic expansion of the geometric function $S(z)$ yields $S(z) \sim \dfrac{4a^2b^2}{z^2}$. Consequently, the magnetic force scales as:

$$F(h) \sim \frac{96}{\pi}\frac{\mu_0 M^2 a^2 b^2 \delta^2}{h^4}. \tag{13}$$

This h^{-4} dependence is a hallmark of dipole-dipole interactions. When the separation h between the rails significantly exceeds their dimensions (a, b, δ), the system effectively behaves as a pair of interacting point dipoles.

Furthermore, as detailed in Appendix B, Eq. (10) can be integrated with respect to z to express the magnetic lift as a potential force. Defining the interaction potential in this manner is particularly advantageous for the subsequent discussion on the stability of the levitating magnet.

IV. RESULTS AND DISCUSSION

IV.1. Experimental Validation and Effective Magnetization

To validate the analytical model, the equilibrium gap h was determined for each total mass by solving the implicit relation (2). We numerically extracted $h(m)$ using the closed-form expression for the magnetic force derived above in eqs. (10) to (12). A single effective strength parameter, $P_{\text{mag}} = \mu_0 M^2/\pi$, was treated as a fitting constant to provide the best agreement with the experimental data.

The results are shown in Fig. 4. The fitted solution reproduces the non-linear decrease of levitation height with increasing mass across the central operating range of the experiment, confirming that the dipole-layer model captures the dominant magnetostatic interaction within experimental uncertainty.

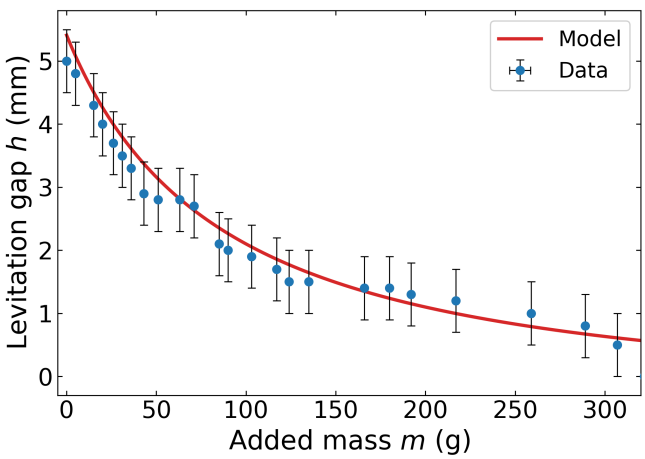

Figure 4. Levitation gap as a function of added mass. Experimental data points (blue circles) with error bars show measured gaps, and the fitted theoretical curve (red line) represents the analytical model derived from magnetic dipole theory.

From Eq. (2), the predicted added mass at a measured gap h_i is

$$m_{\text{pred}}(h_i;\, P_{\text{mag}}) = \frac{2P_{\text{mag}}\,|\Delta_\delta^{(2)}\mathcal{S}(h_i)|}{g} - m_0. \tag{14}$$

The single free parameter P_{mag} is determined by least-squares minimisation of

$$\chi^2 = \sum_{i=1}^{N_{\text{data}}} \left[\frac{m_{\text{pred}}(h_i;\, P_{\text{mag}}) - m_i}{\sigma_m}\right]^2, \tag{15}$$

where $\sigma_m = 1\,\text{g}$ is the mass uncertainty and $N_{\text{data}} = 24$ (the contact point $h = 0$ is excluded).

The best-fit value is $P_{\text{mag}} = (2.5 \pm 0.3) \times 10^4$ Pa. From this result, the effective magnetization for each rail pair is determined as

$$M = \sqrt{\frac{P_{\text{mag}}\pi}{\mu_0}}\,. \tag{16}$$

Because $M \propto P_{\text{mag}}^{1/2}$, the relative uncertainty propagates as $\sigma_M/M = \sigma_{P_{mag}}/(2P_{\text{mag}})$, yielding $M = (2.5 \pm 0.1) \times 10^5\,\text{A/m}$. This value was determined using a bootstrap resampling procedure (5 000 iterations), where the experimental coordinates (h_i, m_i) were perturbed within their measurement tolerances ($\pm 0.5\,\text{mm}$ and $\pm 1\,\text{g}$, respectively) before each re-fit of P_{mag}. This approach captures the combined effect of both experimental error sources, specifically accounting for the dominant role of the gap measurement uncertainty.

The extracted magnetization $M \approx 2.5 \times 10^5\,\text{A/m}$ is consistent with values reported for bonded Nd–Fe–B powders[7] and is physically reasonable for commercially available magnetic tape that has been repolarized into a monopolar configuration.

IV.2. Application to a real-scale Maglev system

The lifting force derived in Sec. III.6 scales linearly with the number N_{pairs} of interacting magnetic elements and quadratically with the magnetization, i.e., $F \propto N_{\text{pairs}} M^2$. This scaling relation enables direct extrapolation of the model to full-scale systems.

To evaluate the implications of this model, we consider a configuration based on the Transrapid TR08 vehicle described by Sobotka[10]. The train consists of three sections with a total mass $m_0 = 188.5$ t and operates at a nominal levitation gap of approximately $h = 10$ mm. The levitation system comprises 46 electromagnets distributed along the vehicle, each of length approximately 0.5 m.

In order to apply the present repulsive magnetostatic model, we construct an equivalent guideway composed of discrete interaction modules. We model the system as a set of uniformly magnetized rectangular prisms of width $2a = 0.12$ m, length $2b = 1.0$ m, and thickness $\delta = 0.03$ m. For a train of length $L = 79.7$ m, this yields $N_{\text{pairs}} = 158$ interacting elements when accounting for both sides of the vehicle.

The total lifting force at the operating gap is then given by

$$F_{\text{total}} = N_{\text{pairs}} \frac{\mu_0 M^2}{\pi} |\Delta_\delta^{(2)}\mathcal{S}(h)|, \tag{17}$$

where $\Delta_\delta^{(2)}\mathcal{S}(h)$ is defined in Eq. eqs. (11) and (12).

Given the constitutive relation $B_r = \mu_0 M$, the total force scales quadratically with the remanence, $F_{\text{total}} \propto B_r^2$.

To estimate passenger capacity, we decompose the total mass into an effective empty vehicle mass and a passenger contribution. Using the reported total mass

188.5 t together with a nominal passenger capacity of approximately 574 individuals, we estimate an empty mass $m_\emptyset \approx 142.6$ t assuming 80 kg per passenger.

The model predicts a minimum remanent flux density $B_r \approx 1.188$ T to levitate the empty vehicle. However, increasing the magnet strength leads to a quadratic increase in lifting capacity. For $B_r \approx 1.366$ T, the model predicts a total lifting capacity matching the full loaded mass, corresponding to a payload of approximately 46 t, or about 574 passengers, matching the reported nominal capacity of the Transrapid TR08.

For higher values of the remanent flux density, the predicted payload increases significantly. For example, for $B_r \approx 1.45$–1.48 T, the model yields payload capacities of 70–80 t, corresponding to approximately 870–990 passengers.

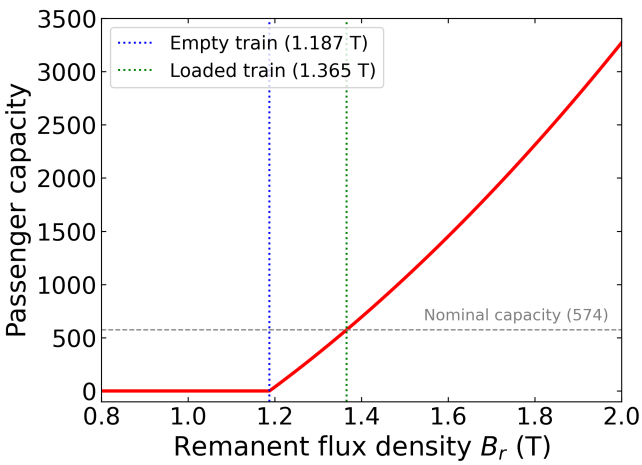

Figure 5. Variation of the train passenger capacity with the magnetic field density of the rail pair B_r for a Transrapid TR08 train.

These results, illustrated by Fig. 5 highlight the dominant role of magnet strength and interaction area in determining levitation capacity. While real Transrapid systems employ actively controlled electromagnetic suspension rather than permanent magnets, this simplified model provides a transparent estimate of the scaling behavior and performance limits achievable in magnetostatic levitation systems.

IV.3. Accuracy and limitations

The goodness-of-fit is assessed via the coefficient of determination:

$$R^2 = 1 - \frac{\sum_i (h_i - h_{\mathrm{pred},i})^2}{\sum_i (h_i - \bar{h})^2}\,, \tag{18}$$

where $h_{\mathrm{pred},i}$ is the gap height obtained by numerically inverting the equilibrium condition in Eq. (2) for each mass m_i. The model yields $R^2 = 0.96$, indicating strong overall agreement with the experimental data.

As a rigorous check of the analytical derivation, we verified that the force computed from Eq. (10) is mathematically equivalent to the classical result of Akoun and Yonnet[8] for the interaction between two cuboidal prisms. For identical geometric and magnetization parameters, the two expressions coincide to machine precision across the entire range of gap heights (1 mm $\leq h \leq$ 100 mm), confirming the formal correctness of the closed-form solutions derived in Appendix B.

Despite this mathematical consistency, the experimental data exhibit systematic physical deviations from the ideal model in four specific regimes:

(i) Small added mass ($m \ll m_0$). In this limit, the equilibrium condition is well-approximated by a first-order Taylor expansion around the unloaded state h_0. This leads to the linear relation:

$$h(m) \approx h_0 - \left(\frac{g}{k(h_0)}\right) m, \tag{19}$$

where $k(h_0) = -\left.\frac{dF}{dh}\right|_{h_0}$ represents the local vertical stiffness. The experimentally observed near-linear decrease in height at small m reflects this first-order behavior of the nonlinear force law.

Notably, while the model captures the slope effectively, the best-fit parameter P_{mag}—determined by a global optimization—leads to a slight systematic departure from the experimental data at $m = 0$. This residual offset suggests that the idealized cuboid model slightly over- or under-estimates the static levitation height h_0 in favor of a better overall fit for higher mass loads.

(ii) Large added mass ($m > 0.30$ kg).

In this regime, the experimental levitation height $h(m)$ drops more abruptly than the model predicts, with the magnets *sticking* at a threshold mass significantly lower than the theoretical limit for $h = 0$. While the analytical expression in Eq. (10) is mathematically valid for all $h > 0$, its physical accuracy diminishes at close proximity. The observed rapid collapse likely stems from the breakdown of the uniform magnetization assumption: at small separations, the strong local fields can induce partial demagnetization or surface-level non-uniformities not captured by the constant-M model. Furthermore, experimental factors such as minor angular misalignments or surface roughness become critical as $h \to 0$, creating a *snap-to-contact* effect where attractive lateral forces or friction overcome the diminishing vertical stiffness.

(iii) Long-rail limit $a \ll b$. In the experimental configuration, the magnetic strips are much longer than they are wide ($a \ll b$). In this limit, the force in the central region of the track is dominated by the longitudinal span, and the end-effects along the y-direction become subleading. Consequently, the magnetic field exhibits approximate translational invariance along the track, allowing the three-dimensional interaction of the rectangular rails to be modeled via a two-dimensional strip approximation.

Practically, this symmetry explains why the measured levitation height is highly sensitive to the lateral align-

ment and the vertical gap h, yet remains nearly invariant with respect to the longitudinal position of the magnet, *provided* it is sufficiently far from the rail ends. Conversely, measurements taken near the boundaries reveal the neglected end-field contributions as an increase in the scatter of $h(m)$. This spatial dependence motivates the protocol of averaging measurements over multiple longitudinal positions to mitigate edge-induced fluctuations.

(iv) Theoretical contact limit ($h \to 0^+$). A critical consistency check for any magnetostatic model is the behavior of the force as the gap vanishes. In models where magnets are idealized as infinitely thin sheets without thickness regularization, field expressions can develop non-physical singularities in the immediate vicinity of the source. In the present derivation, however, the magnetic layer has a finite thickness δ, and the interaction is governed by the second-order finite difference operator $\Delta_\delta^{(2)}$. This construction effectively regularizes the near-field contribution, ensuring a finite limiting force as $h \to 0^+$.

Physically, the force must remain bounded because the magnetization M is finite and the total field energy is well-defined for a magnetized region of non-zero volume. As the gap h becomes small relative to the thickness δ, the force increases rapidly but eventually saturates to a value determined by the aspect ratio a/b and the thickness. While the physical system typically experiences *snap-to-contact* before reaching this limit (see regime ii), the mathematical convergence of Eq. (10) at $h = 0$ confirms that the closed-form expression is robust and free of numerical singularities across all theoretical separation distances.

IV.4. Vertical Stability and Potential Energy

To assess the local vertical stability of the levitation configuration, we examine the total potential energy of the system as a function of the gap h. This energy comprises a repulsive magnetic potential $U_{\text{mag}}(h)$, which decays as the gap increases, and a gravitational contribution $m_0 gh$, which increases linearly with the gap. Figure 6 illustrates these individual contributions and their resultant sum for the unloaded train ($m = 0$).

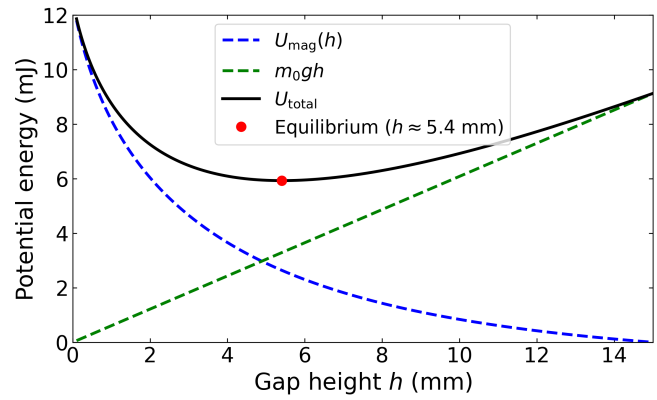

Figure 6. Potential energy as a function of gap height for the empty train. The magnetic potential $U_{\text{mag}}(h)$ (dashed blue) and gravitational potential $m_0 gh$ (dashed green) combine to produce a total potential U_{total} (solid black) with a well-defined minimum at $h \approx 5.4$ mm.

The total potential energy exhibits a well-defined minimum at $h \approx 5.4$ mm, in good agreement with the measured equilibrium gap of 5.0 ± 0.5 mm for the unloaded train. The existence of this stable equilibrium point requires a positive *vertical stiffness*,

$$k(h) \equiv -\frac{dF}{dh} = \frac{d^2U}{dh^2} > 0, \tag{20}$$

which implies that the magnetic force $F_{\text{mag}}(h)$ must decrease monotonically with the gap.

Small vertical perturbations around the equilibrium position generate linear restoring forces: an upward displacement (increasing h) causes the magnetic repulsion to drop below the weight of the train, resulting in a net downward force; conversely, a downward displacement (decreasing h) increases the repulsion, pushing the train back toward its equilibrium position. In this one-dimensional sense, the system is locally stable against vertical fluctuations.

IV.5. Relation to Earnshaw's Theorem

While the vertical force gradient demonstrates a local restoring effect, this does not imply unconditional three-dimensional stability. According to Earnshaw's theorem, a static configuration of permanent magnets cannot achieve a stable equilibrium in all degrees of freedom through purely magnetostatic forces[4]. Specifically, if a point in space is a stable minimum for one direction ($\nabla \cdot \mathbf{F} = 0$), the divergence of the force field dictates that there must be at least one other direction along which the equilibrium is unstable.

In our experimental configuration, the observed vertical stability ($dF_z/dh < 0$) is necessarily compensated by lateral instability. The *levitation* is made possible only because lateral and rotational degrees of freedom are mechanically constrained by the guideway geometry. Thus,

the potential well shown in Fig. 6 characterizes a 1D slice of a 3D saddle-point potential, where the rails provide the physical boundary conditions required to bypass the limitations imposed by Earnshaw's theorem.

IV.6. Comparison with EMS Systems

Industrial high-speed maglev systems, such as the Transrapid, typically employ *electromagnetic suspension* (EMS). In these configurations, vehicle-mounted electromagnets attract ferromagnetic rails from below. Unlike the repulsive system studied here, the magnetic force in an EMS system is attractive and increases as the air gap decreases.

For such an attractive configuration, the force gradient is positive:

$$\frac{\partial F_{\text{mag}}}{\partial h} > 0, \tag{21}$$

which corresponds to a negative vertical stiffness, $k = -\partial F/\partial h < 0$. This state represents an unstable equilibrium: a small upward displacement increases the attractive force, while a small downward displacement decreases it, leading to *runaway* behavior where the magnets either collide with the rail or fall away. Consequently, EMS systems are inherently open-loop unstable and require high-frequency active feedback control of the coil currents to maintain a constant levitation gap.

In contrast, the repulsive permanent-magnet configuration analyzed in this work exhibits a passive, positive vertical stiffness over the entire experimental range. While it does not provide full three-dimensional magnetic stability due to the constraints of Earnshaw's theorem, it offers a self-restoring vertical equilibrium that requires no external power or control logic.

AUTHOR DECLARATIONS

The authors have no conflicts to disclose.

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Appendix A: Experimental Data

Table I. Experimental data for the mass (m) and levitation height (h) of the Maglev train model. The uncertainties are ± 1 g for m and ± 0.5 mm for h.

m (g)	0	5	15	20	26	31	36	43	51	63
h (mm)	5.0	4.8	4.3	4.0	3.7	3.5	3.3	2.9	2.8	2.8
m (g)	71	85	90	103	117	124	135	166	180	192
h (mm)	2.7	2.1	2.0	1.9	1.7	1.5	1.5	1.4	1.4	1.3
m (g)	217	259	289	307	322					
h (mm)	1.2	1.0	0.8	0.5	0.0					

Appendix B: Analytical Derivation of the Force Induced by the Dipole Moment

In this section, we derive the expression for the magnetic force $F(z)$ acting on a Maglev train system, modeled as layers of magnetic dipoles.

1. Magnetic field generated by a rail

We model each rail as a continuous distribution of magnetic dipole moments. Let M be the volume density of dipole moments, i.e., the magnetization of the rail, oriented along the z-axis. The infinitesimal dipole moment

of an element of the rail is defined as:

$$d\boldsymbol{\mu} = M\, dx'\, dy'\, dz'\hat{\mathbf{z}},$$

The vector potential element, $d\mathbf{A}(\mathbf{r})$ (Eq. (3)) corresponding to this dipole moment is expressed in Cartesian coordinates:

$$d\mathbf{A}(\mathbf{r}) = \frac{\mu_0 M}{4\pi} \frac{(-(y-y')\hat{\mathbf{x}} + (x-x')\hat{\mathbf{y}})dx'dy'dz'}{[(x-x')^2 + (y-y')^2 + (z-z')^2]^{3/2}}$$

The total vector potential generated by the lower rail is obtained by integration over the entire surface of the lower rail and over its thickness:

$$\mathbf{A}(\mathbf{r}) = B_0 \int_{Z_1}^{Z_2} \int_{X_1}^{X_2} \int_{Y_1}^{Y_2} \frac{(-Y\hat{\mathbf{x}} + X\hat{\mathbf{y}})dXdYdZ}{[X^2+Y^2+Z^2]^{3/2}}$$

where $B_0 = \mu_0 M/(4\pi)$ and we used the change of variable $X = x - x'$, $Y = y - y'$, $Z = z - z'$, leading to integration boundaries $X_1 = x - a$, $X_2 = x + a$, $Y_1 = y - b$, $Y_2 = y + b$, $Z_1 = z$ and $Z_2 = z + \delta$.

The integrals can be evaluated by successive integration over variables Y, Z, X (or equivalently X, Z, Y).

To streamline subsequent expressions, we first introduce the notational conventions for the radial distances: $R(X,Y,Z) = \sqrt{X^2+Y^2+Z^2}$. The definite integral $\mathcal{K}(X,Z)$, which yields a result with dimensions of inverse length, is defined as:

$$\mathcal{K}(X,Z) = \int_{Y_1}^{Y_2} \frac{-Y\,dY}{R^{3/2}(X,Y,Z)} = \sum_{j=1}^{2} \frac{(-1)^j}{R(X,Y_j,Z)}.$$

Integrating $\mathcal{K}(X,Z)$ with respect to Z yields the dimensionless definite integral $\phi(X) = \int_z^{z+\delta} \mathcal{K}(X,Z)dZ$:

$$\phi(X) = \sum_{j,k=1}^{2} (-1)^{j+k} \sinh^{-1}\left(\frac{Z_k}{(X^2+Y_j^2)^{1/2}}\right).$$

The third stage of integration proceeds by defining the indefinite integral $\mathcal{L}(u,v,w)$. In this expression, u is the integration variable, while v and w are treated as parameters (or constant values).

$$\mathcal{L}(u,v,w) = \int \sinh^{-1}\left(\frac{w}{\sqrt{u^2+v^2}}\right) du.$$

This integral has the following known closed-form analytical solution:

$$\mathcal{L}(u,v,w) = -v \arctan\left(\frac{wu}{vR(u,v,w)}\right) + u \sinh^{-1}\left(\frac{w}{\sqrt{u^2+v^2}}\right) + w \sinh^{-1}\left(\frac{u}{\sqrt{v^2+w^2}}\right).$$

The components of the vector potential $\mathbf{A}(\mathbf{r}) = A_x(x,y,z)\hat{\mathbf{x}} + A_y(x,y,z)\hat{\mathbf{y}}$ directly follow:

$$\begin{aligned} A_x =& B_0 \int_{X_1}^{X_2} \phi_z(X)dX \\ =& B_0 \sum_{i=1}^{2}\sum_{j=1}^{2}\sum_{k=1}^{2} s_{ijk}\mathcal{L}(X_i,Y_j,Z_k), \end{aligned}$$

where $s_{ijk} = (-1)^{i+j+k}$. We find A_y by swapping the roles of X and Y in the integration process:

$$A_y = -B_0 \sum_{i=1}^{2}\sum_{j=1}^{2}\sum_{k=1}^{2} s_{ijk}\mathcal{L}(Y_j,X_i,Z_k)$$

The magnetic field $\mathbf{B}$ is derived from the vector potential $\mathbf{A}$ using the relation: $\mathbf{B} = \nabla \times \mathbf{A}$. Since our analysis focuses on levitation, only the z-component, $B_z = \partial_x A_y - \partial_y A_x$ is relevant. Both $\partial_x A_y$ and $\partial_y A_x$ involve the partial derivative of $\mathcal{L}(u,v,w)$ with respect to the second variable v:

$$\partial_v \mathcal{L}(u,v,w) = -\arctan\left(\frac{wu}{vR(u,v,w)}\right).$$

Using the addition formula

$$\arctan\left(\frac{wu}{vR}\right) + \arctan\left(\frac{wv}{uR}\right) = \arctan\left(\frac{wR}{uv}\right)$$

to combine the two arctan that appear in $\partial_x A_y$ and $\partial_y A_x$, we can express the axial component of the magnetic field in the same form as for a rectangular (cuboid) solenoid[9]:

$$B_z = B_0 \sum_{(i,j,k)=1}^{2} s_{ijk} \arctan\left(\frac{Z_k R(X_i,Y_j,Z_k)}{X_i Y_j}\right).$$

2. Force on the upper rail

The upper rail, placed at a height z, consists of a layer of dipole moments aligned downward:

$$d\boldsymbol{\mu} = -M\, dx\, dy\, dz\hat{\mathbf{z}}.$$

The force on each dipole in the presence of a magnetic field is given by:

$$d\mathbf{F} = \nabla(d\boldsymbol{\mu} \cdot \mathbf{B}).$$

The force component along the z-axis is then expressed as

$$dF_z = -M\partial_z B_z\, dx\, dy\, dz.$$

To find the total force on both upper rails by both lower rails, we integrate over the volume of the upper rail:

$$F_z(h) = -M \int_{-a}^{a} dx \int_{-b}^{b} dy \int_{h}^{h+\delta} dz \partial_z B_z(x,y,z)$$
$$= -M(\Phi(h+\delta) - \Phi(h))$$

where $\Phi(z) = \int_{-a}^{a} dx \int_{-b}^{b} dy B_z(x,y,z)$ denotes the flux of the magnetic field generated by the lower rail through the rectangular surface of the upper rail at z. Using the Stokes theorem and the relation $\mathbf{B} = \nabla \times \mathbf{A}$, we find

$$\Phi(z) = \int_{\mathcal{C}} \mathbf{A}(x,y,z) \cdot (dx\,\hat{\mathbf{x}} + dy\,\hat{\mathbf{y}}) \tag{B1}$$
$$= -\int_{-a}^{a} [A_x]_{-b}^{b} dx + \int_{-b}^{b} [A_y]_{-a}^{a} dy \tag{B2}$$

Calculating the integrands using the fact that $\mathcal{L}_0(u,v,w)$ is an even function of v, we find:

$$[A_x]_{-b}^{b} = 2B_0 \sum_{i,k=1}^{2} (-1)^{i+k} \left(\mathcal{L}_0(X_i, 2b, Z_k) - \mathcal{L}_0(X_i, 0, Z_k)\right)$$
$$[A_y]_{-a}^{a} = -2B_0 \sum_{j,k=1}^{2} (-1)^{j+k} \left(\mathcal{L}_0(Y_j, 2a, Z_k) - \mathcal{L}_0(Y_j, 0, Z_k)\right)$$

Therefore, both terms in (B2) require integrating the function $\mathcal{L}_0(u,v,w)$ with respect to its first variable. We define the integral $\mathcal{A}(u,v,w)$, which has units of a surface area and can be expressed by means of functions $R(u,v,w) = (u^2+v^2+w^2)^{1/2}$ and $\rho(u,v) = (u^2+v^2)^{1/2}$:

$$\mathcal{A}(u,v,w) = \int^{u} \mathcal{L}_0(u',v,w)\,du'$$
$$= -uv \arctan\left(\frac{wu}{vR}\right) + wu \ln(R+u) - \frac{wR}{2}$$
$$+ \frac{u^2-v^2}{2} \ln\left(\frac{R+w}{\rho}\right) - \frac{uw}{2} \ln(v^2+w^2).$$

Since the terms in Eq. (B2) are symmetric with respect to x and y, we define the symmetrized geometric kernel $\mathcal{G}(u,v,w) = \mathcal{A}(u,v,w) + \mathcal{A}(v,u,w)$:

$$\mathcal{G}(u,v,w) = uv\left[\arctan\left(\frac{uv}{wR}\right) - \frac{\pi}{2}\right] - wR$$
$$+ w\left[u \ln\left(\frac{R+u}{\sqrt{v^2+w^2}}\right) + v \ln\left(\frac{R+v}{\sqrt{u^2+w^2}}\right)\right]. \tag{B3}$$

The total flux contribution from a rectangular cross-section is then captured by the surface function $\mathcal{S}(z)$, evaluated as the inclusion-exclusion sum over the rail dimensions. Explicitly:

$$\mathcal{S}(z) = \mathcal{G}(2a,2b,z) - \mathcal{G}(2a,0,z) - \mathcal{G}(0,2b,z) + \mathcal{G}(0,0,z). \tag{B4}$$

With this notation, the magnetic flux through the secondary rail simplifies to:

$$\Phi(z) = -4B_0\left[\mathcal{S}(z+\delta) - \mathcal{S}(z)\right]. \tag{B5}$$

Evaluating the function $\mathcal{S}(z)$ from Eq. (B4) yields the explicit form presented in Eq. (12) of the main text.

The total vertical force is then obtained by the second-order finite difference of this surface function over the magnet thickness δ:

$$F(h) = 4MB_0\Delta_\delta^2 \mathcal{S}(h), \tag{B6}$$

where $\Delta_\delta^2 \mathcal{S}(z) = \mathcal{S}(z+2\delta) - 2\mathcal{S}(z+\delta) + \mathcal{S}(z)$. This operator naturally emerges from the interaction of the two magnetic double-layers.

3. Potential energy

To find the total energy $U(z)$, we integrate over the volume of the upper rail to sum the energies of the dipole elements $-B_z d\mu = M dx dy dz \hat{z}$:

$$U(z) = M \int_{z}^{z+\delta} \left(\int_{-a}^{a} \int_{-b}^{b} B_z(x,y,z')dxdy\right) dz'.$$

This integral can be expressed by integrating the magnetic flux $\Phi(z')$ over the magnet thickness δ:

$$U(z) = M \int_{z}^{z+\delta} \Phi(z')dz'$$
$$= -\frac{\mu_0 M^2}{\pi} \int_{z}^{z+\delta} \left[\mathcal{S}(z'+\delta) - \mathcal{S}(z')\right] dz',$$

which, by defining $\mathcal{T}(z) = \int \mathcal{S}(z')dz'$ as the primitive of the geometric surface function $\mathcal{S}(z)$, reduces to:

$$U(z) = -\frac{\mu_0 M^2}{\pi} \Delta_\delta^{(2)} \mathcal{T}(z). \tag{B7}$$

Here, the operator $\Delta_\delta^{(2)}$ is the second-order forward difference acting on the vertical coordinate, defined in Eq. (11). The geometric potential function $\mathcal{T}(z)$ is evaluated by summing the potential kernel $\mathcal{P}(u,v,z)$ over the magnet dimensions:

$$\mathcal{T}(z) = \mathcal{P}(2a,2b,z) - \mathcal{P}(2a,0,z) - \mathcal{P}(0,2b,z) + \mathcal{P}(0,0,z), \tag{B8}$$

where the potential kernel $\mathcal{P}(u,v,w)$ is given by:

$$\mathcal{P}(u,v,w) = uvw\left[\arctan\left(\frac{uv}{wR}\right) - \frac{\pi}{2}\right] - \frac{3w^2+u^2+v^2}{6}R$$
$$+ \frac{w^2u}{2} \ln\left(\frac{R+u}{\sqrt{v^2+w^2}}\right) + \frac{w^2v}{2} \ln\left(\frac{R+v}{\sqrt{u^2+w^2}}\right)$$
$$+ \frac{u(u^2-3v^2)}{6} \ln(R+u) + \frac{v(v^2-3u^2)}{6} \ln(R+v)$$
$$+ \frac{uv(u+v)}{2} \ln(R+w),$$

and $R \equiv R(u,v,w) = \sqrt{u^2+v^2+w^2}$.